

Data Mining Unit 2 - Unit 2

Data Mining

Unit-2

* Mining frequent Patterns :-

* Frequent patterns are patterns that appear frequently in a dataset. Set of all such items is called frequent itemset.

* It was first proposed by Agarwal, Imielinski and Swami.

* Sequential frequent pattern :- Buying a digital camera and then memory card if it occurs frequently is called frequent sequential pattern.

* Market Basket Analysis :-

Frequent itemset mining leads to discovery of association and correlation among itemset in large transactional dataset.

The process of analyzing customer habits by finding association between items and customer is placed in the shopping basket.

$C_1 \rightarrow$ Milk, Bread, Cereals

$C_2 \rightarrow$ Milk, Bread, Sugar, Eggs

$C_3 \rightarrow$ Milk, Bread, butter

$C_4 \rightarrow$ Sugar, Egg.

Itemset $\rightarrow \{ \text{Milk, Bread, Cereal, Sugar, Egg, Butter} \}$

Frequent Itemset $\rightarrow \{ \text{Milk, Bread, Sugar, Egg} \}$

* Association between any two item is represented in the form of association rules. For example, the information that customer who ~~put~~ buy computer also buy antivirus at same time is represented in association rule as
Computer $\rightarrow$ antivirus [support = 2%, Confidence = 66%]

In General: $A \rightarrow B$ [support, Confidence]

Support = $A \cup B$ [A & B together]
Confidence = A/B [Bought A also bought B]

* The frequent itemset of k item is denoted by L_k .

* Threshold :- Minimum support & confidence above which association can be said interesting.

* Formula

Let $I = \{I_1, I_2, \dots, I_n\}$ be a set of items. Let D be a set of database transaction where each transaction T is a set of items such that $T \subseteq I$. Each transaction is associated with an identifier TID . Let A be set of items where A

transaction T is said to contain A iff $A \subseteq T$.

An association rule is implication of form
 $A \rightarrow B$ where $A \subseteq I, B \subseteq I, A \cap B = \emptyset$

$$\text{Support}(A \rightarrow B) = P(A \cup B)$$

$$\text{Confidence}(A \rightarrow B) = P(A|B)$$

Rules that satisfy both minimum support & confidence are called strong.

* Steps:-

Association rule mining can be viewed as 2 step process:-

- (i) Find all ^{frequent} subset with min-support count.
- (ii) Generate Association rule from frequent itemset.

* Problem

Drawback :- It often generate huge number of itemset satisfying minimum support threshold.
For example, a frequent set of length 100 contains

$\binom{100}{1}$ frequent itemset of 1 item

$\binom{100}{2}$ frequent itemset of 2 item.

and so on

$$\Rightarrow \binom{100}{1} + \binom{100}{2} + \binom{100}{3} + \dots + \binom{100}{100} = 2^{100} - 1 \approx 1.27 \times 10^{30}$$

* Solution.

This is too huge number of itemset for any computer to compute or store. To overcome this difficulty we introduce the concept of ~~frequent~~ closed frequent itemset and maximal frequent itemset.

* An itemset X is closed in a data set S if there exist no proper super-itemset Y such that Y contains same support count as X .

* An itemset X is closed frequent itemset in a set S if X is both closed & frequent.

* An itemset X is a maximal frequent itemset in a set S if X is frequent and there exist no super-itemset Y such that $X \subset Y$ and Y is frequent in S .

* APRIORI Algorithm :- Finding frequent

Itemset	Using Candidate Generation
TID	List of Items
T100	I_1, I_2, I_5
T200	I_2, I_4
T300	I_2, I_3
T400	I_1, I_2, I_4
T500	I_1, I_3
T600	I_2, I_5
T700	I_1, I_5
T800	I_1, I_2, I_3, I_5
T900	I_1, I_2, I_3

Minimum-Support = 2.

⇒ C_1

Itemset	Support-Count
$\{I_1\}$	6
$\{I_2\}$	7
$\{I_3\}$	6
$\{I_4\}$	2
$\{I_5\}$	2

L_1

Itemset	Support-Count
$\{I_1\}$	6
$\{I_2\}$	7
$\{I_3\}$	6
$\{I_4\}$	2
$\{I_5\}$	2

→

Now, we will generate C_2 by joining $L_k \bowtie L_k$
 This Join is possible when we have $k-1$ item is common.

C_2 :

Itemset	Support-Count
$\{I_1, I_2\}$	4
$\{I_2, I_3\}$	4
$\{I_1, I_4\}$	1
$\{I_1, I_5\}$	2
$\{I_2, I_3\}$	4
$\{I_2, I_4\}$	2
$\{I_2, I_5\}$	2
$\{I_3, I_4\}$	0
$\{I_3, I_5\}$	1
$\{I_4, I_5\}$	0

→

Itemset	Sup Count
$\{I_1, I_2\}$	4
$\{I_1, I_3\}$	4
$\{I_1, I_5\}$	2
$\{I_2, I_3\}$	4
$\{I_2, I_4\}$	2
$\{I_2, I_5\}$	2

Here pruning is performed
 (Removal of set whose $\text{Sup-count} < \text{min-Sup}$)

Now, we generate C_3 with $L_2 \bowtie L_2$.

$C_3 :-$

Itemset	Support-Count
$\{I_1, I_2, I_3\}$	2
$\{I_1, I_2, I_5\}$	2
$\{I_1, I_2, I_4\}$	1
$\{I_1, I_3, I_5\}$	1
$\{I_2, I_3, I_4\}$	0
$\{I_2, I_3, I_5\}$	1
$\{I_2, I_4, I_5\}$	0

on joining

$L_3 :-$

Itemset	Support-Count
$\{I_1, I_2, I_3\}$	2
$\{I_1, I_2, I_5\}$	2

On again performing $L_3 \bowtie L_3$ to get C_4 .

Itemset	Support-Count
$\{I_1, I_2, I_3, I_5\}$	1

$L_4 :-$

Q Why it is called Apriori?
 $\Rightarrow$ Because we are utilizing prior information at each step.

Apriori Property :- All non-empty subset of a frequent itemset must also be frequent.

This method is proving a property is Apriori property. This property belongs to special category of property called antimonotone in sense that if a set cannot pass a test, all its subset can also not pass it.

Problem :- We basically have to scan entire database again and again. (frequent scanning).

Generating Association Rules :-

$$\text{Confidence} = P(B/A) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

① Suppose the data contain frequent itemset $I = \{I_1, I_2, I_5\}$. What are the association rules that can be generated from I ? Threshold = 75%

$$\Rightarrow \times \{I_1, I_2\} \rightarrow \{I_5\} = \frac{\text{Support}(A \cup B)}{\text{Support}(A)} = \frac{2}{4} = 50\%$$

$$\{I_1, I_5\} \rightarrow \{I_2\} = \frac{2}{2} = 100\%$$

$$\{I_2, I_5\} \rightarrow \{I_1\} = \frac{2}{2} = 100\%$$

$$\times \{I_1\} \rightarrow \{I_2, I_5\} = 2/6 = 33.33\%$$

$$\times \{I_2\} \rightarrow \{I_1, I_5\} = 2/7 \approx 29\%$$

$$\{I_5\} \rightarrow \{I_1, I_2\} = 2/2 = 100\%$$

$\Rightarrow$ 2nd, 3rd & 6th rule are output.

Improving the efficiency of Apriori Algo:-

1. Hashing technique :- A hash based technique can be used to reduce the size of candidate k -itemsets.

For example :-

$$I_1 \rightarrow 1$$

$$I_2 \rightarrow 2$$

$$I_3 \rightarrow 3$$

$$I_4 \rightarrow 4$$

$$I_5 \rightarrow 5$$

For hash function, $h(u, y) = (10 \times (\text{order of } u) + (\text{order of } y)) \bmod N$

Index	0	1	2	3	4	5	6
Count	1	2	3	2	3	3	2
Content	$\{I_4, I_5\}$	$\{I_1, I_2\}$ $\{I_2, I_3\}$	$\{I_1, I_2\}$ $\{I_3, I_5\}$ $\{I_4, I_5\}$	$\{I_5, I_4\}$ $\{I_1, I_2\}$	$\{I_3, I_5\}$ $\{I_2, I_5\}$ $\{I_4, I_5\}$	$\{I_1, I_2\}$ $\{I_2, I_3\}$ I_3, I_4	$\{I_1, I_2\}$

$$h(I_1, I_2) = (10 \times (1) + (2)) \bmod 7$$

$$= 12 \bmod 7$$

$$= 5$$

If min support is 3. Then, itemset at index 0, 1, 3, 6 can be removed and only item at index 2, 4, 5 will be considered.

2. Transaction Reduction :- It means reducing the no. of ~~least~~ transaction scanned in future iteration.

$$I_1 \rightarrow 6 \quad I_2 \rightarrow 7 \quad I_3 \rightarrow 6 \quad I_4 \rightarrow 2, \quad I_5 \rightarrow 1$$

for min sup of 3, we can remove I_4 & I_5 by transaction reduction.

3 Partitioning :- It involves partitioning the data to find candidate items.

			I_1	I_2	I_3	I_4	I_5
$I_1 = 1$	$I_1 = 1, I_2 = 1, I_4 = 1,$	T_{100}	1	0	0	0	1
$I_2 = 3$	$I_5 = 1, \{I_1, I_5\} = 1$	T_{200}	0	1	0	1	0
$I_3 = 3$	$\{I_2, I_4\} = 1$						
$I_4 = 3$	$I_4 = 1, I_5 = 1, I_2 = 1$	T_{300}	0	0	0	1	1
$I_5 = 2$	$I_3 = 1, \{I_4, I_5\} = 1$	T_{400}	0	1	1	0	0
$\{I_1, I_5\} = 1$	$\{I_2, I_3\} = 1$						
$\{I_2, I_4\} = 2$	$I_5 = 1, I_2 = 1, I_3 = 1, I_4 = 1$	T_{500}	0	0	0	0	1
$\{I_4, I_5\} = 1$	$\{I_2, I_3, I_4\} = 1$	T_{600}	0	1	1	1	0
$\{I_2, I_3\} = 2$	$\{I_2, I_3\} = 1, \{I_3, I_4\} = 1$						
$\{I_3, I_4\} = 1$	$\{I_2, I_4\} = 1$						
$\{I_1, I_3, I_4\} = 1$							

Global

Local

It consist of 2 phase.

- ① Partition of data & find local frequent pattern
- ② Consolidate global frequent pattern.

4 Sampling :- It involves mining on a subset of a given data. The basic idea is to pick a sample from the data and search for frequent items in sample instead of searching entire data.

Size of sample should be that it ~~should~~ can be stored in memory. Because if this does not happen then we will have to store in secondary memory in which time wastage will be more.

FP Growth Algorithm :-

$T_1 : \{E, K, M, N, O, Y\}$

$T_2 : \{D, E, K, N, O, Y\}$

$T_3 : \{A, E, K, M\}$

$T_4 : \{C, K, M, U, Y\}$

$T_5 : \{C, E, I, K, O, O\}$

Min Support 3

Step 1 → Generate the frequency Table.

Item	Frequency.
A	1
C	2
D	1
E	4
I	1
K	5
M	3
N	2
O	3
R	1
U	1
Y	3

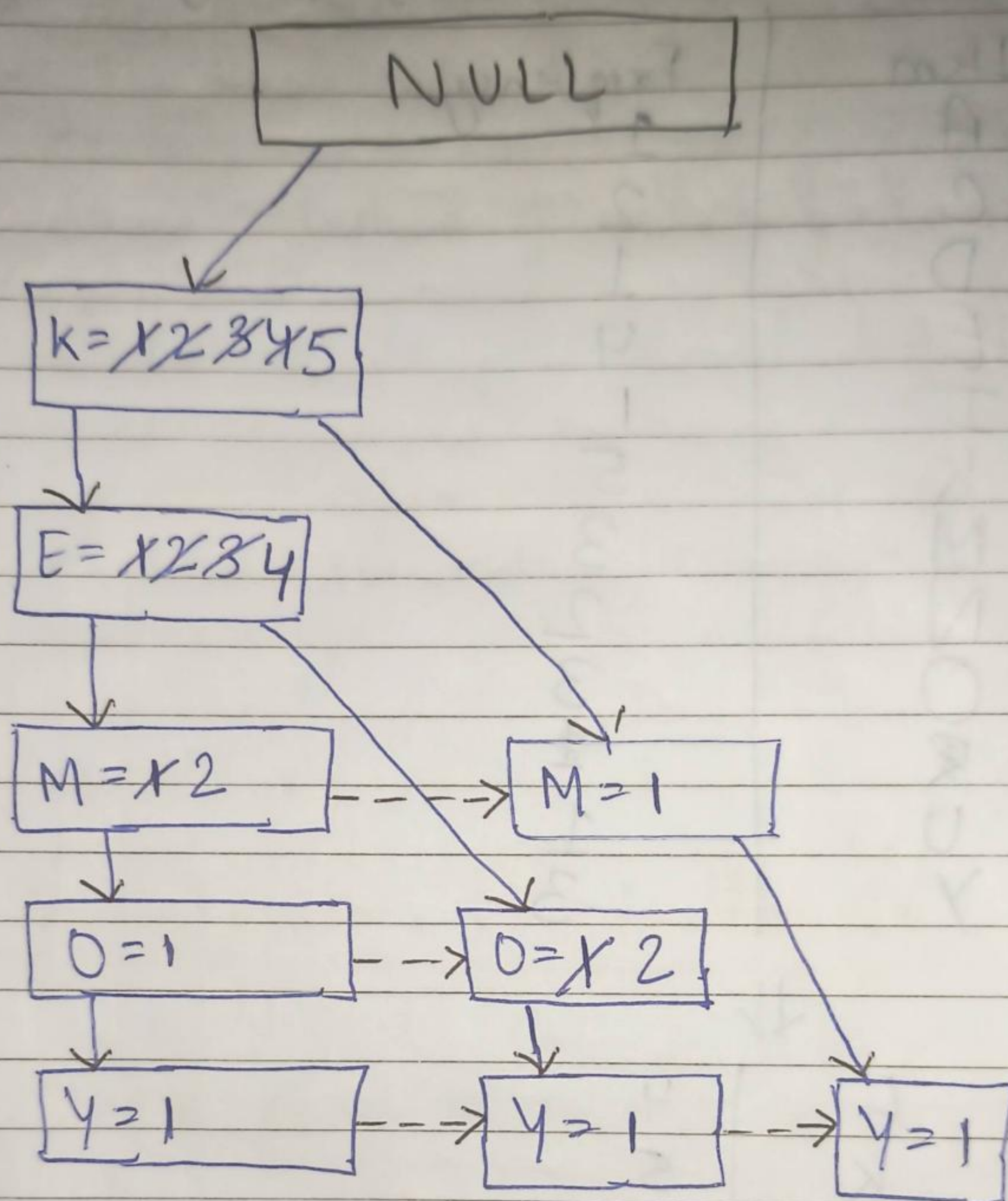
↓

E	4
K	5
M	3
O	3
Y	3

Ordered $\{K=5, E=4, M=3, O=3, Y=3\}$

TID	Item Bought	(Ordered) frequent Item
T ₁	{E, K, M, N, O, Y}	{K, E, M, O, Y}
T ₂	{D, E, K, N, O, Y}	{K, E, O, Y}
T ₃	{A, E, K, M}	{K, E, M}
T ₄	{C, K, M, U, Y}	{K, M, Y}
T ₅	{C, E, I, K, O, O}	{K, E, O}

Step 2 : Draw frequent Pattern Tree



Condition Pattern Base:-

$$Y = \{ \{K, E, M, O; 1\}, \{K, E, O; 1\}, \{K, M; 1\} \}$$

$$O = \{ \{K, E, M; 1\}, \{K, E; 2\} \}$$

$$M = \{ \{K, E; 2\}, \{K; 1\} \}$$

$$E = \{ \{K; 4\} \}$$

$$K = \{ \}$$

Conditional frequent Pattern Tree

Y: {K: 3}

O: {K, E: 3}

M: {K: 3}

E: {K: 4}

Frequent Pattern Generated :-

Y: {K, Y: 3}

O: {K, O: 3}, {E, O: 3}, {K, E, O: 2}

M: {K, M: 3}

E: {K, E: 4}

Benefits :- (i) No need to have Candidate selection after each iteration.
(ii) Complexity Reduce :- No need to scan whole database which was taking $O(n^2)$ time. we just need to scan trie which uses Binary Search $O(\log n)$ time.

* Mining Frequent itemset using Vertical Data format :-

Consider TID and Data set table of Apriori example. That table was based on ~~Trans~~ Transaction. Vertical Data involves Table according to items.

Items	TID Set
I_1	$T_{100}, T_{400}, T_{500}, T_{700}, T_{800}, T_{900}$
I_2	$T_{100}, T_{200}, T_{300}, T_{400}, T_{600}, T_{800}, T_{900}$
I_3	$T_{300}, T_{500}, T_{600}, T_{700}, T_{800}, T_{900}$
I_4	T_{200}, T_{400}
I_5	T_{100}, T_{800}

Itemset	TID Set	Count
$\{I_1, I_2\}$	$T_{100}, T_{400}, T_{800}, T_{900}$	4
$\{I_1, I_3\}$	$T_{500}, T_{700}, T_{800}, T_{900}$	4
X $\{I_1, I_4\}$	T_{400}	1
$\{I_1, I_5\}$	T_{100}, T_{800}	2
$\{I_2, I_3\}$	$T_{300}, T_{600}, T_{800}, T_{900}$	4
$\{I_2, I_4\}$	T_{200}, T_{400}	2
$\{I_2, I_5\}$	T_{100}, T_{800}	2
X $\{I_3, I_4\}$		0
X $\{I_3, I_5\}$	T_{800}	1
X $\{I_4, I_5\}$		0

Item Set	TID	Count
$\{I_1, I_2, I_3\}$	T_{800}, T_{900}	2
$\{I_1, I_2, I_5\}$	T_{800}, T_{900}	2
X $\{I_1, I_2, I_4\}$		
$\{I_1, I_3, I_5\}$		
$\{I_2, I_3, I_4\}$		
$\{I_1, I_3, I_4\}$		
$\{I_2, I_4, I_5\}$		
$\{I_1\}$		
X I_1, I_2, I_3, I_5		0

Support, Confidence :- Strong vs Weak Rule

* Strong $\rightarrow$ If Support & Confidence both greater than min-supp & min-Confidence respectively.

* Weak $\rightarrow$ Otherwise

Q 10,000 transaction.
 6,000 Computer game purchase
 7,500 video game purchase
 4,000 Both game purchase & video

min-supp = 30%

min-Confidence = 60%

Sol $\rightarrow$ buy(x, game) $\rightarrow$ buy(x, video).

$$\begin{aligned}\text{Support} &= P(A \cup B) \\ &= \frac{4000}{10000} \times 100\% \\ &= 40\%.\end{aligned}$$

$$\begin{aligned}\text{Confidence} &= P(A/B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)} \\ &= \frac{4000}{6000} \times 100\% \\ &= 66.66\%\end{aligned}$$

Both greater than its respective threshold

$\Rightarrow$ Strong rule.

However this rule is misleading because probability of purchasing video is 75% which is even larger than confidence (66%). In fact, they ~~are~~ are negatively associated because purchase of one item actually decreases the likelihood of purchasing other.

So, to solve this problem of misleading there must be some other factor in rules alongside support & confidence.

And this is where the concept of correlation came up which is called correlation rules of form

Now, correlation can be found in multiple ways.

→ $A \rightarrow B$ [Support, Confidence, Correlation]

1) Lift :- Lift is simple correlation measure that is given as follows :-

$$\text{Lift} = \frac{P(A \cup B)}{P(A) P(B)}$$

If value of lift $< 1 \rightarrow$ negatively correlated

If value of lift $= 1 \rightarrow$ Events are independent

If value of lift $> 1 \rightarrow$ positively correlated.

Occurrence of one $\Rightarrow$ occurrence of other.

For above example,

$$\text{Lift}(\text{game, video}) = \frac{\frac{4000}{10000}}{\frac{6000}{10000} \times \frac{2500}{10000}} = 0.89$$

$$\text{Lift} < 1$$

So, there is negative correlation between them.

2) Correlation analysis using χ^2 test

	video	<u>video</u>	
game	4000	2000	6000
<u>game</u>	3500	500	4000
	7500	2500	10000

$$\text{Expected (4000)} = \frac{6000 \times 7500}{10000} = 4500$$

$$E(2000) = \frac{2500 \times 6000}{10000} = 1500$$

$$E(3500) = \frac{7500 \times 4000}{10000} = 3000$$

$$E(500) = \frac{2500 \times 4000}{10000} = 1000$$

	game video	<u>video</u>
video game	4000 (4500)	2000 (1500)
<u>game</u>	3500 (3000)	500 (1000)

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

$$\begin{aligned}
 &= \frac{500^2}{4500} + \frac{500^2}{3000} + \frac{500^2}{1500} + \frac{500^2}{1000} \\
 &= 55.55 + 166.67 + 250 + 83.33 \\
 &= 555.55 \\
 &= 555.56
 \end{aligned}$$

Because value of χ^2 is greater than 1 and observed value of slot (game, video) = 4000 which is less than expected value of 4500. So, it is negatively correlated.

$$\begin{aligned}
 \underline{\underline{3.}} \text{ All-Confidence} &= \frac{\text{Support}(A \cup B)}{\max(\text{Support}(A), \text{Support}(B))} \\
 &= \min(P(B|A), P(A|B))
 \end{aligned}$$

Kulczynski Measure :-

$$\frac{1}{2} (P(A|B) + P(B|A))$$

$$\underline{\underline{4.}} \text{ Cosine Measure :- } \frac{\text{Support}(A \cup B)}{\sqrt{\text{Support}(A) \times \text{Support}(B)}}$$

$$\begin{aligned}
 \text{Max-Confidence} &= \frac{\max(P(B|A), P(A|B))}{\text{Support}(A \cup B)} \\
 &= \frac{\text{Support}(A \cup B)}{\min(\text{Support}(A), \text{Support}(B))}
 \end{aligned}$$

For our Example of game & video :-

$$\text{all-corr (game, video)} = \frac{4000}{7500} \\ = 0.53$$

$$\text{Cosine} = \frac{4000}{\sqrt{7500 \times 6000}} \\ = 0.596 \\ = 0.60$$

Based on rule that it is greater than 0.5
So, it is +vely correlated in allcorr & cosine.
Since it was 0-vely correlated in Lift &
 χ^2 . So, good way is to first find it
by all-corr & cosine. If Then if they
are weakly correlated then only other
analysis can be done.